

理论物理笔记

Theoretical Physics Notes

Yee

一份简明的物理学手册编纂计划

Theoretical Physics Notes

First Edition

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Preface

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Chapter 1 向量

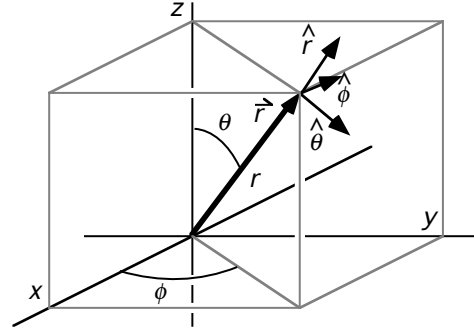
Spherical Coordinates

Transforms

The forward and reverse coordinate transformations are

$$\begin{aligned} r &= \sqrt{x^2 + y^2 + z^2} & x &= r \sin \theta \cos \phi \\ \theta &= \arctan\left(\sqrt{x^2 + y^2}, z\right) & y &= r \sin \theta \sin \phi \\ \phi &= \arctan(y, x) & z &= r \cos \theta \end{aligned}$$

where we *formally* take advantage of the *two argument* arctan function to eliminate quadrant confusion.



Unit Vectors

The unit vectors in the spherical coordinate system are functions of position. It is convenient to express them in terms of the *spherical* coordinates and the unit vectors of the *rectangular* coordinate system which are *not* themselves functions of position.

$$\begin{aligned} \hat{r} &= \frac{\vec{r}}{r} = \frac{x\hat{x} + y\hat{y} + z\hat{z}}{r} = \hat{x} \sin \theta \cos \phi + \hat{y} \sin \theta \sin \phi + \hat{z} \cos \theta \\ \hat{\phi} &= \frac{\hat{z} \times \hat{r}}{\sin \theta} = -\hat{x} \sin \phi + \hat{y} \cos \phi \\ \hat{\theta} &= \hat{\phi} \times \hat{r} = \hat{x} \cos \theta \cos \phi + \hat{y} \cos \theta \sin \phi - \hat{z} \sin \theta \end{aligned}$$

Variations of unit vectors with the coordinates

Using the expressions obtained above it is easy to derive the following handy relationships:

$$\begin{aligned} \frac{\partial \hat{r}}{\partial r} &= 0 \\ \frac{\partial \hat{r}}{\partial \theta} &= \hat{x} \cos \theta \cos \phi + \hat{y} \cos \theta \sin \phi - \hat{z} \sin \theta = \hat{\theta} \\ \frac{\partial \hat{r}}{\partial \phi} &= -\hat{x} \sin \theta \sin \phi + \hat{y} \sin \theta \cos \phi = (-\hat{x} \sin \phi + \hat{y} \cos \phi) \sin \theta = \hat{\phi} \sin \theta \\ \frac{\partial \hat{\phi}}{\partial r} &= 0 \\ \frac{\partial \hat{\phi}}{\partial \theta} &= 0 \\ \frac{\partial \hat{\phi}}{\partial \phi} &= -\hat{x} \cos \phi - \hat{y} \sin \phi = -(\hat{r} \sin \theta + \hat{\theta} \cos \theta) \\ \frac{\partial \hat{\theta}}{\partial r} &= 0 \\ \frac{\partial \hat{\theta}}{\partial \theta} &= -\hat{x} \sin \theta \cos \phi - \hat{y} \sin \theta \sin \phi - \hat{z} \cos \theta = -\hat{r} \\ \frac{\partial \hat{\theta}}{\partial \phi} &= -\hat{x} \cos \theta \sin \phi + \hat{y} \cos \theta \cos \phi = \hat{\phi} \cos \theta \end{aligned}$$

Path increment

We will have many uses for the path increment $d\vec{r}$ expressed in spherical coordinates:

$$\begin{aligned} d\vec{r} &= d(r\hat{r}) = \hat{r}dr + r d\hat{r} = \hat{r}dr + r \left(\frac{\partial \hat{r}}{\partial r} dr + \frac{\partial \hat{r}}{\partial \theta} d\theta + \frac{\partial \hat{r}}{\partial \phi} d\phi \right) \\ &= \hat{r}dr + \hat{\theta}r d\theta + \hat{\phi}r \sin \theta d\phi \end{aligned}$$

Time derivatives of the unit vectors

We will also have many uses for the time derivatives of the unit vectors expressed in spherical coordinates:

$$\begin{aligned} \dot{\hat{r}} &= \frac{\partial \hat{r}}{\partial r} \dot{r} + \frac{\partial \hat{r}}{\partial \theta} \dot{\theta} + \frac{\partial \hat{r}}{\partial \phi} \dot{\phi} = \hat{\theta}\dot{\theta} + \hat{\phi}\dot{\phi} \sin \theta \\ \dot{\hat{\theta}} &= \frac{\partial \hat{\theta}}{\partial r} \dot{r} + \frac{\partial \hat{\theta}}{\partial \theta} \dot{\theta} + \frac{\partial \hat{\theta}}{\partial \phi} \dot{\phi} = -\hat{r}\dot{\theta} + \hat{\phi}\dot{\phi} \cos \theta \\ \dot{\hat{\phi}} &= \frac{\partial \hat{\phi}}{\partial r} \dot{r} + \frac{\partial \hat{\phi}}{\partial \theta} \dot{\theta} + \frac{\partial \hat{\phi}}{\partial \phi} \dot{\phi} = -(\hat{r} \sin \theta + \hat{\theta} \cos \theta)\dot{\phi} \end{aligned}$$

Velocity and Acceleration

The velocity and acceleration of a particle may be expressed in spherical coordinates by taking into account the associated rates of change in the unit vectors:

$$\begin{aligned} \vec{v} &= \dot{\vec{r}} = \dot{r}\hat{r} + r\dot{\hat{r}} \\ \vec{v} &= \dot{r}\hat{r} + \hat{\theta}r\dot{\theta} + \hat{\phi}r\dot{\phi} \sin \theta \\ \vec{a} &= \dot{\vec{v}} = \dot{\hat{r}}\dot{r} + \hat{r}\ddot{r} + \dot{\hat{\theta}}r\dot{\theta} + \hat{\theta}r\ddot{\theta} + \dot{\hat{\theta}}\dot{r} + \hat{\theta}\dot{r}\dot{\theta} + \dot{\hat{\phi}}r\dot{\phi} \sin \theta + \hat{\phi}r\ddot{\phi} \sin \theta + \hat{\phi}r\dot{\phi}\dot{\theta} \cos \theta + \hat{\phi}r\dot{\phi}\dot{\theta} \cos \theta \\ &= (\hat{\theta}\dot{\theta} + \hat{\phi}\dot{\phi} \sin \theta)\dot{r} + \hat{r}\ddot{r} + (-\hat{r}\dot{\theta} + \hat{\phi}\dot{\phi} \cos \theta)r\dot{\theta} + \hat{\theta}\dot{r}\dot{\theta} + \hat{\theta}r\ddot{\theta} \\ &\quad + \left[-(\hat{r} \sin \theta + \hat{\theta} \cos \theta)\dot{\phi} \right] r\dot{\phi} \sin \theta + \hat{\phi}r\ddot{\phi} \sin \theta + \hat{\phi}r\dot{\phi}\dot{\theta} \cos \theta + \hat{\phi}r\dot{\phi}\dot{\theta} \cos \theta \\ \vec{a} &= \hat{r}(\ddot{r} - r\dot{\theta}^2 - r\dot{\phi}^2 \sin^2 \theta) + \hat{\theta}(r\ddot{\theta} + 2\dot{r}\dot{\theta} - r\dot{\phi}^2 \sin \theta \cos \theta) + \hat{\phi}(r\ddot{\phi} \sin \theta + 2r\dot{\theta}\dot{\phi} \cos \theta + 2\dot{r}\dot{\phi} \sin \theta) \end{aligned}$$

The del operator from the definition of the gradient

Any (static) scalar field u may be considered to be a function of the spherical coordinates r , θ , and ϕ . The value of u changes by an infinitesimal amount du when the point of observation is changed by $d\vec{r}$. That change may be determined from the partial derivatives as

$$du = \frac{\partial u}{\partial r} dr + \frac{\partial u}{\partial \theta} d\theta + \frac{\partial u}{\partial \phi} d\phi.$$

But we also define the gradient in such a way as to obtain the result

$$du = \vec{\nabla} u \cdot d\vec{r}$$

Therefore,

$$\frac{\partial u}{\partial r} dr + \frac{\partial u}{\partial \theta} d\theta + \frac{\partial u}{\partial \phi} d\phi = \vec{\nabla} u \cdot d\vec{r}$$

or, in spherical coordinates,

$$\frac{\partial u}{\partial r} dr + \frac{\partial u}{\partial \theta} d\theta + \frac{\partial u}{\partial \phi} d\phi = (\vec{\nabla} u)_r dr + (\vec{\nabla} u)_\theta r d\theta + (\vec{\nabla} u)_\phi r \sin \theta d\phi$$

and we demand that this hold for any choice of dr , $d\theta$, and $d\phi$. Thus,

$$(\vec{\nabla}u)_r = \frac{\partial u}{\partial r}, \quad (\vec{\nabla}u)_\theta = \frac{1}{r} \frac{\partial u}{\partial \theta}, \quad (\vec{\nabla}u)_\phi = \frac{1}{r \sin \theta} \frac{\partial u}{\partial \phi},$$

from which we find

$$\boxed{\vec{\nabla} = \hat{r} \frac{\partial}{\partial r} + \frac{\hat{\theta}}{r} \frac{\partial}{\partial \theta} + \frac{\hat{\phi}}{r \sin \theta} \frac{\partial}{\partial \phi}}$$

Divergence

The divergence $\vec{\nabla} \cdot \vec{A}$ is carried out taking into account, once again, that the unit vectors themselves are functions of the coordinates. Thus, we have

$$\vec{\nabla} \cdot \vec{A} = \left(\hat{r} \frac{\partial}{\partial r} + \frac{\hat{\theta}}{r} \frac{\partial}{\partial \theta} + \frac{\hat{\phi}}{r \sin \theta} \frac{\partial}{\partial \phi} \right) \cdot (A_r \hat{r} + A_\theta \hat{\theta} + A_\phi \hat{\phi})$$

where the derivatives must be taken *before* the dot product so that

$$\begin{aligned} \vec{\nabla} \cdot \vec{A} &= \left(\hat{r} \frac{\partial}{\partial r} + \frac{\hat{\theta}}{r} \frac{\partial}{\partial \theta} + \frac{\hat{\phi}}{r \sin \theta} \frac{\partial}{\partial \phi} \right) \cdot \vec{A} \\ &= \hat{r} \cdot \frac{\partial \vec{A}}{\partial r} + \frac{\hat{\theta}}{r} \cdot \frac{\partial \vec{A}}{\partial \theta} + \frac{\hat{\phi}}{r \sin \theta} \cdot \frac{\partial \vec{A}}{\partial \phi} \\ &= \hat{r} \cdot \left(\frac{\partial A_r}{\partial r} \hat{r} + \frac{\partial A_\theta}{\partial r} \hat{\theta} + \frac{\partial A_\phi}{\partial r} \hat{\phi} + A_r \frac{\partial \hat{r}}{\partial r} + A_\theta \frac{\partial \hat{\theta}}{\partial r} + A_\phi \frac{\partial \hat{\phi}}{\partial r} \right) \\ &\quad + \frac{\hat{\theta}}{r} \cdot \left(\frac{\partial A_r}{\partial \theta} \hat{r} + \frac{\partial A_\theta}{\partial \theta} \hat{\theta} + \frac{\partial A_\phi}{\partial \theta} \hat{\phi} + A_r \frac{\partial \hat{r}}{\partial \theta} + A_\theta \frac{\partial \hat{\theta}}{\partial \theta} + A_\phi \frac{\partial \hat{\phi}}{\partial \theta} \right) \\ &\quad + \frac{\hat{\phi}}{r \sin \theta} \cdot \left(\frac{\partial A_r}{\partial \phi} \hat{r} + \frac{\partial A_\theta}{\partial \phi} \hat{\theta} + \frac{\partial A_\phi}{\partial \phi} \hat{\phi} + A_r \frac{\partial \hat{r}}{\partial \phi} + A_\theta \frac{\partial \hat{\theta}}{\partial \phi} + A_\phi \frac{\partial \hat{\phi}}{\partial \phi} \right) \end{aligned}$$

With the help of the partial derivatives previously obtained, we find

$$\begin{aligned} \vec{\nabla} \cdot \vec{A} &= \hat{r} \cdot \left(\frac{\partial A_r}{\partial r} \hat{r} + \frac{\partial A_\theta}{\partial r} \hat{\theta} + \frac{\partial A_\phi}{\partial r} \hat{\phi} + 0 + 0 + 0 \right) \\ &\quad + \frac{\hat{\theta}}{r} \cdot \left(\frac{\partial A_r}{\partial \theta} \hat{r} + \frac{\partial A_\theta}{\partial \theta} \hat{\theta} + \frac{\partial A_\phi}{\partial \theta} \hat{\phi} + A_r \hat{\theta} + A_\theta (-\hat{r}) + 0 \right) \\ &\quad + \frac{\hat{\phi}}{r \sin \theta} \cdot \left(\frac{\partial A_r}{\partial \phi} \hat{r} + \frac{\partial A_\theta}{\partial \phi} \hat{\theta} + \frac{\partial A_\phi}{\partial \phi} \hat{\phi} + A_r \sin \theta \hat{\phi} + A_\theta \cos \theta \hat{\phi} + A_\phi \left[-(\hat{r} \sin \theta + \hat{\theta} \cos \theta) \right] \right) \\ &= \left(\frac{\partial A_r}{\partial r} \right) + \left(\frac{1}{r} \frac{\partial A_\theta}{\partial \theta} + \frac{A_r}{r} \right) + \left(\frac{1}{r \sin \theta} \frac{\partial A_\phi}{\partial \phi} + \frac{A_r}{r} + \frac{A_\theta \cos \theta}{r \sin \theta} \right) \\ &= \left(\frac{\partial A_r}{\partial r} + \frac{2A_r}{r} \right) + \left(\frac{1}{r} \frac{\partial A_\theta}{\partial \theta} + \frac{A_\theta \cos \theta}{r \sin \theta} \right) + \frac{1}{r \sin \theta} \frac{\partial A_\phi}{\partial \phi} \\ &\boxed{\vec{\nabla} \cdot \vec{A} = \frac{1}{r^2} \frac{\partial}{\partial r} (r^2 A_r) + \frac{1}{r \sin \theta} \frac{\partial}{\partial \theta} (A_\theta \sin \theta) + \frac{1}{r \sin \theta} \frac{\partial A_\phi}{\partial \phi}} \end{aligned}$$

Curl

The curl $\vec{\nabla} \times \vec{A}$ is also carried out taking into account that the unit vectors themselves are functions of the coordinates. Thus, we have

$$\vec{\nabla} \times \vec{A} = \left(\hat{r} \frac{\partial}{\partial r} + \frac{\hat{\theta}}{r} \frac{\partial}{\partial \theta} + \frac{\hat{\phi}}{r \sin \theta} \frac{\partial}{\partial \phi} \right) \times (A_r \hat{r} + A_\theta \hat{\theta} + A_\phi \hat{\phi})$$

where the derivatives must be taken *before* the cross product so that

$$\begin{aligned} \vec{\nabla} \times \vec{A} &= \left(\hat{r} \frac{\partial}{\partial r} + \frac{\hat{\theta}}{r} \frac{\partial}{\partial \theta} + \frac{\hat{\phi}}{r \sin \theta} \frac{\partial}{\partial \phi} \right) \times \vec{A} \\ &= \hat{r} \times \frac{\partial \vec{A}}{\partial r} + \frac{\hat{\theta}}{r} \times \frac{\partial \vec{A}}{\partial \theta} + \frac{\hat{\phi}}{r \sin \theta} \times \frac{\partial \vec{A}}{\partial \phi} \\ &= \hat{r} \times \left(\frac{\partial A_r}{\partial r} \hat{r} + \frac{\partial A_\theta}{\partial r} \hat{\theta} + \frac{\partial A_\phi}{\partial r} \hat{\phi} + A_r \frac{\partial \hat{r}}{\partial r} + A_\theta \frac{\partial \hat{\theta}}{\partial r} + A_\phi \frac{\partial \hat{\phi}}{\partial r} \right) \\ &\quad + \frac{\hat{\theta}}{r} \times \left(\frac{\partial A_r}{\partial \theta} \hat{r} + \frac{\partial A_\theta}{\partial \theta} \hat{\theta} + \frac{\partial A_\phi}{\partial \theta} \hat{\phi} + A_r \frac{\partial \hat{r}}{\partial \theta} + A_\theta \frac{\partial \hat{\theta}}{\partial \theta} + A_\phi \frac{\partial \hat{\phi}}{\partial \theta} \right) \\ &\quad + \frac{\hat{\phi}}{r \sin \theta} \times \left(\frac{\partial A_r}{\partial \phi} \hat{r} + \frac{\partial A_\theta}{\partial \phi} \hat{\theta} + \frac{\partial A_\phi}{\partial \phi} \hat{\phi} + A_r \frac{\partial \hat{r}}{\partial \phi} + A_\theta \frac{\partial \hat{\theta}}{\partial \phi} + A_\phi \frac{\partial \hat{\phi}}{\partial \phi} \right) \end{aligned}$$

With the help of the partial derivatives previously obtained, we find

$$\begin{aligned} \vec{\nabla} \times \vec{A} &= \hat{r} \times \left(\frac{\partial A_r}{\partial r} \hat{r} + \frac{\partial A_\theta}{\partial r} \hat{\theta} + \frac{\partial A_\phi}{\partial r} \hat{\phi} + 0 + 0 + 0 \right) \\ &\quad + \frac{\hat{\theta}}{r} \times \left(\frac{\partial A_r}{\partial \theta} \hat{r} + \frac{\partial A_\theta}{\partial \theta} \hat{\theta} + \frac{\partial A_\phi}{\partial \theta} \hat{\phi} + A_r \hat{\theta} + A_\theta (-\hat{r}) + 0 \right) \\ &\quad + \frac{\hat{\phi}}{r \sin \theta} \times \left(\frac{\partial A_r}{\partial \phi} \hat{r} + \frac{\partial A_\theta}{\partial \phi} \hat{\theta} + \frac{\partial A_\phi}{\partial \phi} \hat{\phi} + A_r \sin \theta \hat{\phi} + A_\theta \cos \theta \hat{\phi} + A_\phi \left[-(\hat{r} \sin \theta + \hat{\theta} \cos \theta) \right] \right) \\ &= \left(\frac{\partial A_\theta}{\partial r} \hat{\phi} - \frac{\partial A_\phi}{\partial r} \hat{\theta} \right) + \left(-\frac{1}{r} \frac{\partial A_r}{\partial \theta} \hat{\phi} + \frac{1}{r} \frac{\partial A_\phi}{\partial \theta} \hat{r} + \frac{A_\theta}{r} \hat{\phi} \right) \\ &\quad + \left(\frac{1}{r \sin \theta} \frac{\partial A_r}{\partial \phi} \hat{\theta} - \frac{1}{r \sin \theta} \frac{\partial A_\theta}{\partial \phi} \hat{r} - \frac{A_\phi}{r} \hat{\theta} + \frac{A_\phi \cos \theta}{r \sin \theta} \hat{r} \right) \\ &= \hat{r} \left(\frac{1}{r} \frac{\partial A_\phi}{\partial \theta} - \frac{1}{r \sin \theta} \frac{\partial A_\theta}{\partial \phi} + \frac{A_\phi \cos \theta}{r \sin \theta} \right) \\ &\quad + \hat{\theta} \left(-\frac{\partial A_\phi}{\partial r} + \frac{1}{r \sin \theta} \frac{\partial A_r}{\partial \phi} - \frac{A_\phi}{r} \right) \\ &\quad + \hat{\phi} \left(\frac{\partial A_\theta}{\partial r} - \frac{1}{r} \frac{\partial A_r}{\partial \theta} + \frac{A_\theta}{r} \right) \end{aligned}$$

$$\boxed{\vec{\nabla} \times \vec{A} = \frac{\hat{r}}{r \sin \theta} \left[\frac{\partial}{\partial \theta} (A_\phi \sin \theta) - \frac{\partial A_\theta}{\partial \phi} \right] + \frac{\hat{\theta}}{r \sin \theta} \left[\frac{\partial A_r}{\partial \phi} - \sin \theta \frac{\partial}{\partial r} (r A_\phi) \right] + \frac{\hat{\phi}}{r} \left[\frac{\partial}{\partial r} (r A_\theta) - \frac{\partial A_r}{\partial \theta} \right]}$$

Laplacian

The Laplacian is a scalar operator that can be determined from its definition as

$$\begin{aligned}\nabla^2 u &= \vec{\nabla} \cdot (\vec{\nabla} u) = \left(\hat{r} \frac{\partial}{\partial r} + \frac{\hat{\theta}}{r} \frac{\partial}{\partial \theta} + \frac{\hat{\phi}}{r \sin \theta} \frac{\partial}{\partial \phi} \right) \cdot \left(\hat{r} \frac{\partial u}{\partial r} + \frac{\hat{\theta}}{r} \frac{\partial u}{\partial \theta} + \frac{\hat{\phi}}{r \sin \theta} \frac{\partial u}{\partial \phi} \right) \\ &= \hat{r} \cdot \frac{\partial}{\partial r} \left(\hat{r} \frac{\partial u}{\partial r} + \frac{\hat{\theta}}{r} \frac{\partial u}{\partial \theta} + \frac{\hat{\phi}}{r \sin \theta} \frac{\partial u}{\partial \phi} \right) \\ &\quad + \frac{\hat{\theta}}{r} \cdot \frac{\partial}{\partial \theta} \left(\hat{r} \frac{\partial u}{\partial r} + \frac{\hat{\theta}}{r} \frac{\partial u}{\partial \theta} + \frac{\hat{\phi}}{r \sin \theta} \frac{\partial u}{\partial \phi} \right) \\ &\quad + \frac{\hat{\phi}}{r \sin \theta} \cdot \frac{\partial}{\partial \phi} \left(\hat{r} \frac{\partial u}{\partial r} + \frac{\hat{\theta}}{r} \frac{\partial u}{\partial \theta} + \frac{\hat{\phi}}{r \sin \theta} \frac{\partial u}{\partial \phi} \right)\end{aligned}$$

With the help of the partial derivatives previously obtained, we find

$$\begin{aligned}\nabla^2 u &= \hat{r} \cdot \left(\hat{r} \frac{\partial^2 u}{\partial r^2} - \frac{\hat{\theta}}{r^2} \frac{\partial u}{\partial \theta} + \frac{\hat{\theta}}{r} \frac{\partial^2 u}{\partial \theta \partial r} - \frac{\hat{\phi}}{r^2 \sin \theta} \frac{\partial u}{\partial \phi} + \frac{\hat{\phi}}{r \sin \theta} \frac{\partial^2 u}{\partial \phi \partial r} \right) \\ &\quad + \frac{\hat{\theta}}{r} \cdot \left(\hat{\theta} \frac{\partial u}{\partial r} + \hat{r} \frac{\partial^2 u}{\partial r \partial \theta} - \frac{\hat{r}}{r} \frac{\partial u}{\partial \theta} + \frac{\hat{\theta}}{r} \frac{\partial^2 u}{\partial \theta^2} - \frac{\hat{\phi} \cos \theta}{r \sin^2 \theta} \frac{\partial u}{\partial \phi} + \frac{\hat{\phi}}{r \sin \theta} \frac{\partial^2 u}{\partial \phi \partial \theta} \right) \\ &\quad + \frac{\hat{\phi}}{r \sin \theta} \cdot \left(\hat{\phi} \sin \theta \frac{\partial u}{\partial r} + \hat{r} \frac{\partial^2 u}{\partial r \partial \phi} + \frac{\hat{\phi} \cos \theta}{r} \frac{\partial u}{\partial \theta} + \frac{\hat{\theta}}{r} \frac{\partial^2 u}{\partial \theta \partial \phi} - \frac{\hat{r} \sin \theta + \hat{\theta} \cos \theta}{r \sin \theta} \frac{\partial u}{\partial \phi} + \frac{\hat{\phi}}{r \sin \theta} \frac{\partial^2 u}{\partial \phi^2} \right) \\ &= \left(\frac{\partial^2 u}{\partial r^2} \right) + \left(\frac{1}{r} \frac{\partial u}{\partial r} + \frac{1}{r^2} \frac{\partial^2 u}{\partial \theta^2} \right) + \left(\frac{1}{r} \frac{\partial u}{\partial r} + \frac{\cos \theta}{r^2 \sin \theta} \frac{\partial u}{\partial \theta} + \frac{1}{r^2 \sin^2 \theta} \frac{\partial^2 u}{\partial \phi^2} \right) \\ &= \left(\frac{\partial^2 u}{\partial r^2} + \frac{2}{r} \frac{\partial u}{\partial r} \right) + \left(\frac{1}{r^2} \frac{\partial^2 u}{\partial \theta^2} + \frac{\cos \theta}{r^2 \sin \theta} \frac{\partial u}{\partial \theta} \right) + \left(\frac{1}{r^2 \sin^2 \theta} \frac{\partial^2 u}{\partial \phi^2} \right) \\ &= \frac{1}{r^2} \frac{\partial}{\partial r} \left(r^2 \frac{\partial u}{\partial r} \right) + \frac{1}{r^2 \sin \theta} \frac{\partial}{\partial \theta} \left(\sin \theta \frac{\partial u}{\partial \theta} \right) + \frac{1}{r^2 \sin^2 \theta} \frac{\partial^2 u}{\partial \phi^2}\end{aligned}$$

Thus, the Laplacian operator can be written as

$$\boxed{\nabla^2 = \frac{1}{r^2} \frac{\partial}{\partial r} \left(r^2 \frac{\partial}{\partial r} \right) + \frac{1}{r^2 \sin \theta} \frac{\partial}{\partial \theta} \left(\sin \theta \frac{\partial}{\partial \theta} \right) + \frac{1}{r^2 \sin^2 \theta} \frac{\partial^2}{\partial \phi^2}}$$

Chapter 2 特征值问题

2.1 1

Chapter 3 傅里叶变换

3.1 傅里叶级数

魏尔斯特拉斯定理告诉我们任意关于 x 的连续函数都可以展开成 x 的代数多项式。这个结论可以推广到 n 个自变量的情形：

Theorem 3.1.1 (Generalized Stone-Weierstrass Theorem): 如果函数 $f(x_1, x_2, \dots, x_n)$ 是区间 $[a_i, b_i]^n$ 上的连续函数，那么 $f(x_1, x_2, \dots, x_n)$ 可以展开成多项式 $x_1^{k_1} x_2^{k_2} \dots x_n^{k_n}$, k_i 是非负整数。

由 $x_1 = x = r \cos \theta, x_2 = y = r \sin \theta$ ，我们得到了极坐标下展开一个二元函数的方法：

$$f(r, \theta) = \sum_{m,k=0}^{\infty} a_{mk} x^k y^m = \sum_{m,k=0}^{\infty} a_{m,k} r^{m+k} \cos^k \theta \sin^m \theta \quad (3.1)$$

如果我们令 $r = 1$ （或者其他常数）就得到了一个只关于 θ 的函数：

$$f(\theta) = \sum_{n=-\infty}^{\infty} F_n e^{in\theta} = F_0 + \sum_{n=1}^{\infty} (A_n \cos \theta + B_n \sin \theta) \quad (3.2)$$

这是一个周期为 2π 的周期性函数，它可以表示 $[-\pi, \pi]$ 上的连续函数（或者分段连续函数，准确地讲是满足狄利克雷条件的函数）。利用正交性和归一性，不难求出 [1]：

$$F_n = \frac{1}{2\pi} \int_{-\pi}^{\pi} e^{-in\theta} f(\theta) d\theta \quad (3.3)$$

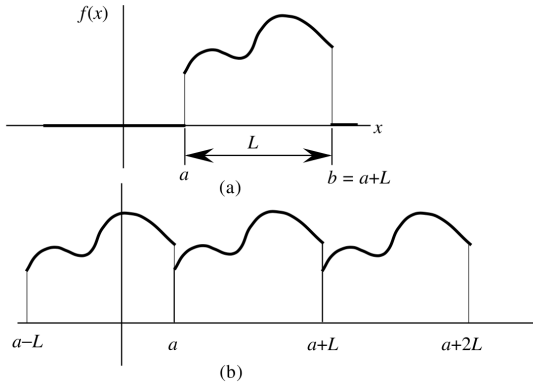
但是，我们感兴趣的函数并不总是定义在 $[-\pi, \pi]$ 上。更一般的，对于定义在 (a, b) 上，周期为 $L=(b-a)$ 的函数 $f(x)$ ，通过换元

$$\theta = \frac{2\pi}{L}(x - a - \frac{L}{2})$$

可以得到

$$f(x) = \sum_{n=-\infty}^{\infty} F_n e^{2\pi i n x / L} \quad (3.4)$$

$$F_n = \frac{1}{L} \int_a^b e^{-2\pi i n x / L} f(x) dx \quad (3.5)$$



(a) The function we want to represent. (b) The Fourier series representation of the function

3.2 傅里叶变换

傅里叶级数可以展开任意周期性函数 $F(x)$, 但是物理学中遇到的大多数函数并不是周期性的。一种简单的想法是使周期 $L \rightarrow \infty$, 此时 $\Delta\omega = \frac{2\pi}{L} \rightarrow 0$ 。[2] 我们直接给出傅里叶变换的定义

$$F[f(x)] = G(\omega) = \int_{-\infty}^{\infty} f(x) e^{-i\omega x} dx \quad (3.6)$$

$$F^{-1}[G(\omega)] = f(x) = \frac{1}{2\pi} \int_{-\infty}^{\infty} G(\omega) e^{i\omega x} d\omega \quad (3.7)$$

我们称公式3.6为函数 $f(x)$ 的傅里叶变换, $F[f(x)] = G(\omega)$ 称为 $f(x)$ 的像函数. 公式3.7称为 $G(\omega)$ 的傅里叶逆变换, $f(x)$ 称为 $G(\omega)$ 的像原函数.

上面的定义可以推广到三维:

$$G(\omega) = \int_{-\infty}^{\infty} f(r) e^{-i\omega \cdot r} dr \quad (3.8)$$

$$f(r) = \frac{1}{2\pi} \int_{-\infty}^{\infty} G(\omega) e^{i\omega \cdot r} d\omega \quad (3.9)$$

其中:

$$\begin{cases} \omega = \omega_1 \mathbf{e}_1 + \omega_2 \mathbf{e}_2 + \omega_3 \mathbf{e}_3 \\ \mathbf{r} = x \mathbf{e}_1 + y \mathbf{e}_2 + z \mathbf{e}_3 \\ d\mathbf{r} = dx dy dz \\ f(\mathbf{r}) = f(x, y, z) \end{cases}$$

3.2.1 傅里叶变换的性质

- 线性性质: $F[\alpha f_1 + \beta f_2] = \alpha F[f_1] + \beta F[f_2]$.
- 延迟性质: $F[e^{i\omega_0 x} f(x)] = G(\omega - \omega_0)$.
- 位移性质: $F[f(x - x_0)] = e^{-i\omega x_0} F[f(x)]$.
- 相似性质: $F[f(ax)] = \frac{1}{|a|} G(\frac{\omega}{a})$.
- 微分性质: $F[f^{(n)}(x)] = (i\omega)^n F[f(x)]$, 正是由于傅里叶变换的微分性质, 引出了求解微分方程的积分变换法.

- 积分性质: $F[\int_{x_0}^x f(\xi)d\xi] = \frac{1}{i\omega}F[f(x)]$.
- 卷积性质: $f_1(x) * f_2(x) = \int_{-\infty}^{\infty} f_1(x)f_2(x-\xi)d\xi$ 称为卷积, 卷积满足交换律、结合律、分配律。对函数和像函数:

$$F[f_1(x) * f_2(x)] = F[f_1(x)] \cdot F[f_2(x)] \quad (\text{卷积定理}) \quad (3.10)$$

$$F[f_1(x) \cdot f_2(x)] = \frac{1}{2\pi} F[f_1(x)] * F[f_2(x)] \quad (3.11)$$

Chapter 4 数学物理方程

4.1 达朗贝尔法

4.2 分离变量法

4.3 积分变换法

Chapter 5 格林函数

Chapter 6 一些物理学中常用的函数

Chapter 7 张量

Chapter 8 物理学中的群论

Chapter 9 经典力学

Chapter 10 量子力学基础

本章主要介绍量子力学中的基本概念和数学工具，包括狄拉克符号、表象理论、角动量、全同粒子等，关于量子力学的诠释和历史，将会在自然哲学部分中介绍。

参考书目：

- 《量子力学导论》，格里菲斯，很经典的入门书籍，有中文版；
- 《量子力学》，科恩-塔诺季，值得收藏的一本书，大师之作，详略得当，附录非常全面；
- 《量子力学原理》，王正行，这本书花费了很多篇幅在狄拉克符号上，适合对此感到困惑的初学者；
- 《高等量子力学》，喀兴林，表象理论和全同粒子介绍的比较详细。

10.1 量子力学的基本假设

1. 描写微观系统状态的数学量是希尔伯特空间 (态空间) 中的矢量 (态矢量 $|\psi\rangle$)，相差一个复数因子的两个矢量，描写同一状态。
2. 每一个可测量的物理量 $\mathcal{A}$ 都可以用希尔伯特空间中的一个厄米算符 A 描述，因此之后我们统一使用大写字母标记，这个算符是一个观察算符；
3. 物理量所能取的值 a_i ，只能是相应算符 A 的本征值之一，因为 A 是厄米的，因此测量结果总是实数；

$$A|a_i\rangle = a_i|a_i\rangle \quad \langle a_i|a_j\rangle = \delta_{ij} \quad (10.1)$$

4. 物理量 A 在状态 $|\psi\rangle$ 中取各值 a_i 的概率，与态矢量 $|\psi\rangle$ 按 A 的归一化本征矢量 $|a_i\rangle$ 的展开式中 $|a_i\rangle$ 的系数 c_i 的复平方成正比。若 $|\psi\rangle$ 是归一化的，那么测量物理量 A 得到 a_i 的概率是 $|c_i|^2$ 。因此可以写出 A 在状态 $|\psi\rangle$ 的平均值 $\langle A \rangle$ ：

$$|\psi\rangle = \sum_i |a_i\rangle c_i = \sum_i |a_i\rangle \langle a_i|\psi\rangle \quad (10.2)$$

$$\langle A \rangle = \frac{\sum_i c_i^* c_i a_i}{\sum_i c_i^* c_i} = \frac{\sum_i \langle \psi|a_i\rangle \langle a_i|\psi\rangle a_i}{1} = \langle \psi|A|\psi\rangle \quad (10.3)$$

5. 微观系统中每个粒子的直角坐标下的位置算符 X_i ($i=1, 2, 3$)，与相应的正则动量算符 P ，有下列对易关系：

$$[X_i, X_j] = 0 \quad [P_i, P_j] = 0 \quad [X_i, P_j] = i\hbar\delta_{ij} \quad \text{其中 } \hbar = \frac{h}{2\pi} \quad (10.4)$$

而不同粒子间的所有算符相互对易 $[A_i, B_j] = 0$ 。

6. 微观系统的状态随时间演化的方式是 (含时) 薛定谔方程:

$$i\hbar \frac{\partial}{\partial t} |\psi\rangle = H|\psi\rangle \quad (10.5)$$

只要给出了初始状态 $|\psi(0)\rangle$ 就足以知道此后任意时刻的状态 $|\psi(t)\rangle$ 。在演化过程中没有任何不确定性, 只在测量时发生不可预测的跃变。

7. 描写全同粒子系统的态矢量, 对于任意一对粒子的对调, 是对称的 (对调前后完全相同) 或反对称的 (对调前后差一个负号), 服从前者的粒子称为玻色子, 服从后者的粒子称为费米子。

10.2 谐振子和矩阵方法

10.3 角动量

Chapter 11 量子力学 II

11.1 全同粒子和二次量子化

11.2 路径积分

11.3 狄拉克方程

Chapter 12 狭义相对论

Chapter 13 固体物理基础

13.1 从波函数到宏观物理量

凝聚态物理的起点是多粒子的薛定谔方程, $\Psi(\{\mathbf{r}_i\}, t)$ 是反对称的多粒子波函数, 简单起见我们暂时不考虑自旋。

$$i\hbar \frac{d\Psi(\{\mathbf{r}_i\}, t)}{dt} = \hat{H}\Psi(\{\mathbf{r}_i\}, t) \quad (13.1)$$

哈密顿算符是位置算符、动量算符（以及时间和系统所处外部环境）的函数，其函数形式与经典力学的哈密顿量相同，包含了系统与相互作用相关的全部信息。

$$\hat{H} = -\frac{\hbar^2}{2m_e} \sum_i \nabla_i^2 - \frac{\hbar^2}{2M_I} \sum_I \nabla_I^2 - \sum_{i,I} \frac{Z_I e^2}{|\mathbf{r}_i - \mathbf{R}_I|} + \frac{1}{2} \sum_{i \neq j} \frac{e^2}{|\mathbf{r}_i - \mathbf{r}_j|} + \frac{1}{2} \sum_{I \neq J} \frac{Z_I Z_J e^2}{|\mathbf{R}_I - \mathbf{R}_J|} \quad (13.2)$$

其中 Z_I 和 M_I 是原子核的电荷数和质量。

13.1.1 电子密度

$$\hat{n}(\mathbf{r}) = \sum_{i=1}^N \delta(\mathbf{r} - \mathbf{r}_i) \quad (13.3)$$

$$n(\mathbf{r}) = \frac{\langle \Psi | \hat{n}(\mathbf{r}) | \Psi \rangle}{\langle \Psi | \Psi \rangle} = N \frac{\int d^3r_2 d^3r_3 \cdots d^3r_N |\Psi(\mathbf{r}, \mathbf{r}_2, \mathbf{r}_3 \cdots \mathbf{r}_N)|^2}{\int d^3r_1 d^3r_2 d^3r_3 \cdots d^3r_N |\Psi(\mathbf{r}_1, \mathbf{r}_2, \mathbf{r}_3 \cdots \mathbf{r}_N)|^2} \quad (13.4)$$

13.1.2 能量

忽略掉原子核的动能，系统的哈密顿量可以分为以下几部分：

$$\hat{H} = \hat{T} + \hat{V}_{\text{ext}} + \hat{V}_{\text{int}} + E_{II} \quad (13.5)$$

它们分别对应于公式电子动能、电子与核的相互作用、电子之间的相互作用、原子核之间的相互作用。 $\hat{H}$ 的期望值也可以拆成这几项：

$$E \equiv \langle \hat{H} \rangle = \frac{\langle \Psi | \hat{H} | \Psi \rangle}{\langle \Psi | \Psi \rangle} = \langle \hat{T} \rangle + \langle \hat{V}_{\text{int}} \rangle + \int d^3r V_{\text{ext}}(\mathbf{r}) n(\mathbf{r}) + E_{II} \quad (13.6)$$

如果将能量 E 看做是波函数 Ψ 的泛函，那么哈密顿量的本征态就是使能量取得极值的鞍点。我们用拉格朗日乘子法处理公式13.6，得到的结果是定态薛定谔方程：

$$\hat{H}|\Psi\rangle = E|\Psi\rangle \quad (13.7)$$

13.2 晶体结构

- 理想的晶体是由相同的**原子团**（注意是原子团而不是原子）在空间中**无限重复**排列构成的，这样的原子团称为**基元**（*basis*）。
- 在研究晶体的结构时，这些原子团可以抽象为几何点，所有点的集合称为**晶格**（*lattice*）。
- 由于晶体具有平移对称性，可以通过平移 $u_1\vec{a}_1 + u_2\vec{a}_2 + u_3\vec{a}_3$ 使每一个格点移动到完全等价的另一个格点。如果以上的操作可以遍历全部格点，那么这个晶格就称为初基晶格， $\vec{a}_i$ 称为**初基平移矢量**（*primitive translation vector*）。基元内原子 j 的中心位置也可以用 $\vec{a}$ 来表示：

$$\mathbf{r}_j = x_j\mathbf{a}_1 + y_j\mathbf{a}_2 + z_j\mathbf{a}_3$$

初基平移矢量的选取不是唯一的。我们往往用初基平移矢量来定义晶轴。

- 体积最小的晶胞称为原胞。初基平移矢量所确定的平行六面体 $(\vec{a}_1 \cdot \vec{a}_2 \times \vec{a}_3)$ 是原胞的一种选取方法，因为初基平移矢量的选取不唯一，原胞选取也不唯一。
- **维格纳-赛茨原胞**：原胞的另一种取法，把一个格点和相邻格点用直线连接起来，做连线的垂直平分线或者垂直平分面，围成的最小体积。

13.3 倒空间

周期性的函数可以展开成傅里叶级数，对于晶体中的波函数：

$$n(\mathbf{r}) = \sum_{\mathbf{G}} n_{\mathbf{G}} e^{i\mathbf{G} \cdot \mathbf{r}} \quad (13.8)$$

$$n_{\mathbf{G}} = \frac{1}{V_c} \int_{cell} n(\vec{r}) e^{-i\mathbf{G} \cdot \mathbf{r}} d\vec{r} \quad (13.9)$$

$\mathbf{G}$ 不是任意的，因为 $n(\mathbf{r})$ 具有平移不变性：

$$n(\mathbf{r} + \vec{T}) = \sum_{\mathbf{G}} n_{\mathbf{G}} e^{i\mathbf{G} \cdot \mathbf{r}} e^{i\mathbf{G} \cdot \mathbf{T}} = n(\mathbf{r}) \quad (13.10)$$

我们定义倒格子的轴矢量：

$$\mathbf{b}_1 = 2\pi \frac{\mathbf{a}_2 \times \mathbf{a}_3}{\mathbf{a}_1 \cdot \mathbf{a}_2 \times \mathbf{a}_3} \quad \mathbf{b}_2 = 2\pi \frac{\mathbf{a}_3 \times \mathbf{a}_1}{\mathbf{a}_2 \cdot \mathbf{a}_3 \times \mathbf{a}_1} \quad \mathbf{b}_3 = 2\pi \frac{\mathbf{a}_1 \times \mathbf{a}_2}{\mathbf{a}_3 \cdot \mathbf{a}_1 \times \mathbf{a}_3} \quad (13.11)$$

倒空间每个初基矢量和实空间的两个轴矢量正交：

$$\mathbf{b}_i \cdot \mathbf{a}_j = 2\pi \delta_{ij} \quad (13.12)$$

当 $\mathbf{G} = v_1\mathbf{b}_1 + v_2\mathbf{b}_2 + v_3\mathbf{b}_3$ (v_i 是整数) 时, $\exp(i\mathbf{G} \cdot \mathbf{T}) = 1$, 因此 $\mathbf{G} = v_1\mathbf{b}_1 + v_2\mathbf{b}_2 + v_3\mathbf{b}_3$ 就是波矢被允许的取值范围。

13.3.1 布里渊区

布里渊区的定义是倒格子空间中的维格纳-赛茨原胞，它的边界是服从衍射条件的波矢 $\mathbf{k}$ ：

$$2\mathbf{k} \cdot \mathbf{G} = G^2 \quad (13.13)$$

倒格子的中央晶胞称为第一布里渊区。

13.4 固体能带理论

13.5 附录: 三维空间的十四种晶格

13.6 附录: 晶体衍射条件

Chapter 14 粒子物理基础

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